

Replicating the paper: Temporal Relational Ranking for Stock Prediction (Feng et al., 2019)

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Abstract

Stock returns are usually modeled from each stock’s own history, although firms linked through an industry, a supply chain, or common owners tend to move together. Feng et al. (2019) close this gap with Relational Stock Ranking (RSR), which passes information between related stocks through a Temporal Graph Convolution and ranks stocks by next-day return, and they report gains over relation-blind baselines. This replication retrains the paper’s relation-blind Rank_LSTM and its industry-relation RSR on the authors’ released NASDAQ data and code, under a matched 50-epoch schedule, and repeats the comparison in a second run. Mean-squared error is indistinguishable between the two models, and the baseline has the higher top-pick reciprocal rank, as in the paper’s own NASDAQ results; the investment return favors RSR in both of our runs, opposite to the paper, but it is too noisy to settle the direction. We are interested in this paper because we are building a multi-agent trading system whose agents each analyze one stock in isolation, and we want to know whether a relation graph adds a signal before wiring one in.

Keywords: Stock Prediction, Learning to Rank, Graph-based Learning

JEL Codes: G17, C45, C53¹

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¹Feng et al. (2019) do not report JEL codes, so we chose these ourselves from the JEL classification system: G17 (Financial Forecasting and Simulation), C45 (Neural Networks and Related Topics), and C53 (Forecasting and Prediction Methods; Simulation Methods).

1 Motivation

A stock’s future return is usually modeled from its own history, which discards information that is easy to observe: companies in the same industry, supply chain, or fund tend to move together, because a shock to one is often a shock to the group. Empirical finance shows that information travels along such links and reaches prices with a delay. Customer and supplier industries cross-predict each other’s returns (Menzly and Ozbas, 2010), news about economically linked firms is priced slowly (Cohen and Frazzini, 2008), and money managers’ commodity-futures positions predict producers’ stock returns the following week (Ho and Lauwers, 2023). Links also shape real outcomes: upstream tariff cuts raise downstream investment (Martin and Otto, 2024), and shared institutional owners change supply-chain relationships (Freeman, 2025). If links carry information, a model that treats each stock as an independent series ignores part of the signal, and the question is whether encoding the links improves prediction.

Feng et al. (2019) answer with Relational Stock Ranking (RSR). A graph of relations between stocks, either shared sector–industry membership or Wikidata facts on ownership, supply, and competition, enters a Temporal Graph Convolution layer on top of an LSTM encoding of each stock’s price history. The model is trained to rank stocks by next-day return rather than to regress it, since a trading strategy cares about relative order more than exact values. The paper compares RSR with a relation-blind LSTM ranker (Rank_LSTM), a graph-regularized regressor (GBR), and a plain graph convolution (GCN) on NASDAQ and NYSE. Its claims are conditional. With industry relations RSR beats Rank_LSTM on NYSE but not on NASDAQ, which the authors attribute to NASDAQ being more volatile and dominated by short-term factors; with Wikidata relations it reaches an investment return ratio of 1.19 against 0.68 for Rank_LSTM on NASDAQ (their Tables 6 and 7).

Other machine-learning work builds on the same idea. Cui et al. (2023) extend pairwise relations to group-wise hypergraphs of industry belonging and fund holding with hierarchical attention, and Jafari and Haratizadeh (2022) apply a graph convolutional network to a stock-relation graph. Evgeniou et al. (2023) use machine learning to show that firm-level return predictability is sparse and heterogeneous, so which firms carry signal for which others is an empirical matter, and Feng et al. (2024) use deep learning to build latent factors from firm characteristics. We set out to reproduce the paper’s NASDAQ, industry-relation comparison between Rank_LSTM and RSR, using the authors’ released data and code. We could obtain the same NASDAQ data and the same relation graph. We could not obtain the authors’ pretrained sequential embedding that RSR starts from, so we exported it from our own baseline run; we did not tune hyperparameters as the authors did; and we did not run

NYSE, Wikidata relations, or the GBR and GCN comparators. A reader learns three things the paper alone does not show: how the two models compare under one fixed, matched schedule; how large run-to-run variation is for each metric; and which defects in the released data a reproducer must find first (Section 2).

Our interest comes from outside this course. We are building a multi-agent system for trading decisions in which each agent analyzes one stock along one dimension (technical, news, risk). No agent sees the relationship between stocks, which is the assumption RSR argues against. Replicating RSR is a first check of whether that signal is real before any such agent is added. This project does not attempt that integration or extend RSR.

1.1 Research question

- **RQ1 (replication).** On the NASDAQ universe with industry relations, does RSR improve next-day return ranking relative to Rank_LSTM, which treats stocks independently?

2 Data

We use the NASDAQ portion of the dataset released with Feng et al. (2019): daily features for 1,026 tickers over 1,246 trading days (January 2013 to December 2017) and a static industry relation tensor. The repository describes its raw pull as thirty years of data on more than 8,000 stocks. The modeling sample is about five years, and only 1,026 stocks pass the paper’s liquidity and history filters. Our data is the authors’ own, so it does not differ from the original paper’s data.

2.1 Dependent and independent variables

Table 1 lists the variables. The response is the next-day return ratio. The predictors are the closing price, normalized by the stock’s maximum over the sample, and its 5-, 10-, 20-, and 30-day moving averages. The model reads the last four days of these five features. The relation graph is a fixed input shared by all dates.

2.2 Data exploration, cleaning and preprocessing

Exploratory analysis (notebook `eda.ipynb`) found three issues.

1. **Missing values are not NaN.** The pipeline (`preprocess/eod.py`) pads missing trading days with -1234 . Left in, the sentinel drives raw skewness to about -23 for every

Table 1: Variables.

Acronym	Description	Type
RET	Response: next-day return ratio, $(p_{t+1} - p_t)/p_t$, which the models predict and rank across stocks	continuous
CLOSE	Closing price divided by the stock’s maximum over 2013–2017	continuous
MA5, MA10, MA20, MA30	5-, 10-, 20-, 30-day moving averages of the normalized close	continuous
IND	Relation graph: $1,026 \times 1,026 \times 97$ one-hot tensor of shared industry membership	binary

feature column. Stripped, it falls to between -0.3 and -0.4 , a mild skew in a bounded ratio. The sentinel marks only 0.17% of ticker-days (2,202 of 1,278,396 rows), yet it dominates any unfiltered statistic. We treat it as missing wherever it appears; because so few rows are affected, we add no missing indicator.

2. **The graph includes non-company tickers.** 156 of 1,026 tickers (15.2%) have industry “n/a”. They are ETFs and index funds (e.g. QQQ, SHV) whose return is a basket average with no industry relation, and 17 of them are isolated in the graph. We keep them, as the paper does, because dropping them changes the graph; the effect is untested.
3. **The graph differs from the paper’s description.** The tensor has 97 industry channels and 5.2% related stock pairs. The paper’s Table 4 reports 112 relation types and 5.00%. We could not account for the gap in types.

Next-day return ratios range from -0.48 to 0.98 , and only 0.05% exceed 0.2 in absolute value. The authors’ ticker filter already caps daily returns near 0.98, we did not trace individual extremes to the source, and we kept them, as the paper does. The features are already scaled by each stock’s maximum price, so we did not rescale.

3 Methodology

3.1 Benchmark model

We use the paper’s own benchmark, Rank-LSTM. A single-layer LSTM reads each stock’s last $S = 4$ days of features and predicts next-day return from the final hidden state, with no information exchanged between stocks. This is the “stocks are independent” model that RSR

is designed to beat. The relational model is RSR with implicit relation modeling (RSR-I in the paper). It takes the baseline’s learned sequence embedding and re-weights it over each stock’s neighbors in the industry graph with a learned, softmax-normalized attention, then combines the propagated and original embeddings for the prediction. The graph is a training-time constant. Both models use the loss

$$\mathcal{L} = \|\hat{\mathbf{r}}_{t+1} - \mathbf{r}_{t+1}\|^2 + \alpha \sum_{i,j} \max(0, -(\hat{r}_{t+1}^i - \hat{r}_{t+1}^j)(r_{t+1}^i - r_{t+1}^j)),$$

a regression term plus a pairwise penalty for stock pairs whose predicted order contradicts the realized order.

3.2 Experimental Design

Figure 1 shows the procedure from raw data to the reported numbers.

Both models share the date split of the paper (756 training, 252 validation, and the remaining test days), optimizer (Adam, learning rate 0.001), seq = 4, unit = 32, and $\alpha = 1$, and both train for 50 epochs. We report test metrics at the epoch with the best validation loss. The paper tuned hyperparameters by grid search on the investment return and averaged five test runs; the repository’s example command for Wikidata relations uses seq = 16 and unit = 64. Our contribution is not a new method but a test the paper does not report: a matched-schedule comparison, repeated in a second independent run (a new baseline run, hence a new embedding, and RSR with a different random seed), so that run-to-run variation is visible.

3.3 Performance measures

We use the paper’s three measures. *MSE* is the mean squared error of predicted return across stocks and test days. *MRR* is the mean over test days of the reciprocal of the true-return rank of the model’s top-ranked stock (in the code, `mrrt`). *IRR* is the cumulative investment return ratio, the sum of the realized next-day returns of the top-ranked stock over the test days. The code reports `bt1`, which equals one plus this sum, so we report $\text{IRR} = \text{bt1} - 1$ to match the paper’s scale. The paper notes that IRR swings widely because swapping the top two stocks on one high-return day changes it greatly; we therefore treat IRR as the noisiest of the three.

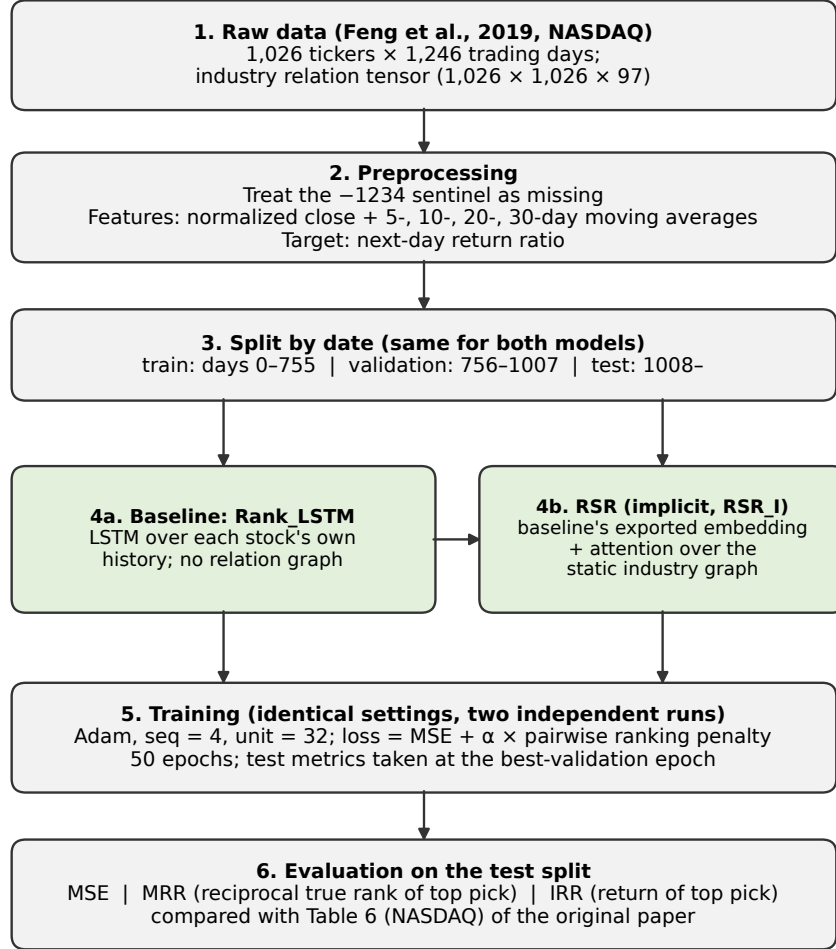


Figure 1: Experimental design, from raw data to reported test-set metrics.

4 Results

Table 2 puts the paper’s NASDAQ industry-relation results beside ours. The paper averages five test runs; we have two independent runs.

Table 2: NASDAQ, industry relations: original paper (Table 6; mean $\pm$ s.d. over five test runs) and this replication (50 epochs, test metrics at best validation epoch). MSE in 10^{-4} .

Source	Model	MSE (10^{-4})	MRR	IRR
Paper	Rank_LSTM	3.79 ± 0.01	0.0417 ± 0.0075	0.68 ± 0.60
Paper	RSR_I	3.80 ± 0.01	0.0317 ± 0.0051	0.23 ± 0.27
Replication, run 1	Rank_LSTM	3.774	0.0488	0.018
Replication, run 1	RSR_I	3.774	0.0276	0.130
Replication, run 2	Rank_LSTM	3.773	0.0322	−0.205
Replication, run 2	RSR_I	3.787	0.0301	0.405

MSE replicates. The two models agree to within 0.4% in both runs, as in the paper (3.79 against 3.80). A relation graph does not lower prediction error here.

MRR replicates in direction. The baseline has the higher MRR in both runs, as in the paper, though the size of the gap varies (0.021 in run 1, 0.002 in run 2, against 0.010 in the paper). The gap in run 2 is well inside the paper’s standard deviations.

IRR does not replicate, but cannot decide the question. RSR has the higher IRR in both runs, the reverse of the paper’s ordering (0.23 against 0.68). Yet the baseline’s IRR moves from 0.018 to −0.205 between our two runs, and RSR’s from 0.130 to 0.405, swings as large as the paper’s own standard deviations (0.60 and 0.27). Two runs cannot establish the sign of a difference this noisy.

So the answer to RQ1 on NASDAQ with industry relations is no for MSE and MRR and unresolved for IRR. This agrees with the paper’s own statement that industry relations help less on NASDAQ. Four things may account for the IRR difference, none of which we can separate with two runs: we did not grid-search hyperparameters on IRR as the authors did; we used two runs, not five; and the embedding was exported from our own baseline run, not the authors’ file, so its training may differ.

5 Conclusion

What survives the replication is the paper’s NASDAQ industry-relation finding in its weaker form: a static industry graph leaves mean-squared error unchanged and does not improve the ranking of the single best stock, both of which the original Table 6 also shows. The

released code and data were enough to reproduce this, once two traps were cleared: the undocumented `-1234` missing-value code, and an embedding file that the repository does not ship.

What a reader should believe less is any claim that RSR raises investment return on NASDAQ from industry relations alone. The sign of our IRR difference is the opposite of the paper’s, and both are small next to run-to-run variation, so neither supports a conclusion. Needed next are more seeds, a hyperparameter search toward the paper’s configuration, and the Wikidata relation, where the paper’s own NASDAQ gain is largest. For the multi-agent system that motivated this project, the practical reading is that a static industry graph is not a reliable source of an extra signal on NASDAQ; whether a richer relation is, remains untested here.

Declaration

Declaration of Pedagogical Purpose

This manuscript is prepared as part of the course project for Machine Learning and FinTech at National Yang Ming Chiao Tung University, instructed by Professor Huei-Wen Teng. The work is intended solely for pedagogical purposes, including learning, discussion, and academic training. It does not represent original research submitted for journal publication, nor should it be considered professional financial advice.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Claude, developed by Anthropic, in order to improve the clarity, precision, and overall quality of the writing and to assist with code and analysis. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the article.

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A Reproduction notes

Training code is the official repository’s `rank_lstm.py` and `relation_rank_lstm.py`, with TensorFlow 1 calls run under TensorFlow 2 and the environment variable `TF_USE_LEGACY_KERAS=1` set; without it the scripts fail at `BasicLSTMCell`. We added a `--save_emb` option to the baseline to export the sequence embedding and an `--epochs` option to both. Run 1 uses the default seed. In run 2 the baseline script has no seed option, so it differs from run 1 through initialization alone, and RSR was run with `--seed 2` on run 2’s embedding. Logs are in `coding/logs/`.